

Machine Learning Integrated Tail Risk Detection: Combining GARCH Volatility Dynamics, Extreme Value Theory, and Gradient Boosting for Crisis Prediction

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Abstract: Traditional Value at Risk models systematically failed during the 2008 financial crisis, underestimating tail risk through static volatility assumptions and backward-looking calibration. This paper introduces a machine learning integrated framework that transforms econometric models, specifically GARCH volatility dynamics and Extreme Value Theory parameters, into engineered features for supervised tail risk prediction. Rather than estimating current VaR values, we reframe the problem as predicting whether future returns will exceed VaR thresholds, enabling integration of multiple risk signals into a unified early warning system. Using daily returns from five major U.S. investment banks during 2005–2010, we train an XGBoost classifier to predict tail events, achieving 60.1% AUC on in-sample data with performance improving to 64.2% during the crisis period. Out-of-sample validation on the COVID-19 market crash achieves 73.5% AUC, representing a 22.3% improvement over in-sample performance despite no exposure to pandemic-era data during training. SHAP analysis reveals that GARCH conditional volatility dominates predictions with 21.8% importance, while novel interaction features capturing volatility and cross-sectional dispersion rank second at 14.1%, detecting systemic stress propagation that isolated models miss. Statistical validation includes Chow tests confirming structural breaks, Kupiec and Christoffersen backtests demonstrating traditional VaR failures, and robustness analysis across threshold specifications. The framework contributes to quantitative risk management by providing an interpretable, regime-adaptive screening tool for risk committees.

Keywords: Tail Risk; GARCH; Extreme Value Theory; Machine Learning; Financial Crises; XGBoost; Regime Detection; Value at Risk

JEL Classification: G32, C53, C58, G01

1. Introduction

The 2008 global financial crisis exposed fundamental weaknesses in quantitative risk management systems. Banks and financial institutions relying on Value at Risk models, regulatory staples for measuring downside exposure, systematically underestimated the probability and magnitude of extreme losses. Post-crisis analysis revealed that these failures stemmed not from implementation errors but from conceptual limitations inherent to standalone risk models. The recognition that traditional VaR methodologies exhibit fundamental limitations has motivated extensive research into alternative approaches, yet a comprehensive framework integrating volatility dynamics, tail modeling, and pattern recognition remains very limited in the existing literature.

Traditional VaR methodologies exhibit four fundamental limitations that became acutely apparent during the crisis. First, static volatility assumptions in historical simulation and

parametric VaR ignore the well-documented phenomenon of volatility clustering, where large price movements beget subsequent large movements. Bollerslev [1] formalized this persistence through the GARCH framework, demonstrating that volatility exhibits strong autocorrelation structures that simple rolling-window estimators cannot capture. During the 2008 crisis, the VIX index surged from approximately 20 to over 80, yet historical simulation VaR using fixed lookback windows responded sluggishly because equal weighting of historical observations dilutes the impact of recent volatility spikes. GARCH models, by contrast, assign exponentially decaying weights that respond rapidly to regime changes, providing volatility forecasts that anticipate near-term risk more accurately than backward-looking averages.

Second, Gaussian distribution assumptions severely underweight the probability of extreme events, contradicting overwhelming empirical evidence of fat tails in financial returns documented extensively since Mandelbrot [2]. The normal distribution implies that a six-standard-deviation event should occur approximately once every four million years, yet financial markets experience such events with disturbing regularity. During October 2008, daily returns exceeding five standard deviations occurred multiple times within a single month. This discrepancy arises because the normal distribution's exponentially decaying tails assign vanishingly small probabilities to extreme outcomes, while empirical return distributions exhibit power-law tail decay where extreme events remain probabilistically meaningful. Extreme Value Theory addresses this limitation through the Generalized Pareto Distribution, which models tail behavior without imposing restrictive assumptions about the complete distribution, allowing the data to reveal tail thickness through the estimated shape parameter.

Third, backward-looking calibration windows cannot anticipate regime shifts, which represent precisely the moments when risk models are most needed. Traditional VaR estimates quantile thresholds based on historical return distributions, implicitly assuming stationarity that fails dramatically during crisis transitions. When markets shift from low-volatility to high-volatility regimes, VaR estimates calibrated on tranquil periods systematically understate risk until sufficient crisis observations enter the estimation window. This delay can span weeks or months, during which risk models provide false assurance precisely when danger is greatest. The fundamental problem is that VaR answers the question "what loss threshold was exceeded with probability p historically?" rather than the forward-looking question "what is the probability that tomorrow's return will exceed a given threshold?"

Fourth, methodological isolation prevents leveraging complementary information across modeling approaches. GARCH captures volatility dynamics through autoregressive variance structures but provides no direct information about tail thickness or the probability of extreme events conditional on current volatility levels. Extreme Value Theory captures tail behavior through the Generalized Pareto Distribution but treats observations as independent draws, ignoring the temporal clustering that characterizes financial crises where extreme events beget subsequent extremes. Machine learning excels at pattern recognition and can discover complex nonlinear relationships, but purely data-driven approaches lack the domain structure that econometric theory provides, risking spurious correlations and poor out-of-sample performance. Each methodology addresses a subset of tail risk dimensions while remaining blind to others, yet practitioners typically apply these tools in isolation rather than synthesizing their complementary strengths.

This paper addresses these limitations by developing an integrated framework that transforms traditional econometric outputs into predictive features for supervised learning. The central insight motivating this approach is that econometric models produce rich intermediate outputs, including conditional volatility estimates, standardized residuals, and tail distribution parameters, that contain predictive information about future tail events beyond what any single output captures in isolation. Rather than treating GARCH volatility forecasts as final products for direct VaR calculation, we treat them as features encoding forward-looking volatility dynamics. Rather than using EVT solely to estimate static tail quantiles, we compute rolling tail parameters that capture time-varying tail thickness evolution. This perspective transforms econometric modeling from an endpoint into a preprocessing stage that structures raw return data into economically meaningful representations suitable for pattern recognition.

We reframe VaR measurement as a binary classification problem, predicting whether future returns will exceed VaR thresholds rather than estimating threshold values directly. This reformulation offers several advantages over traditional quantile estimation. Classification enables direct optimization of tail event detection through appropriate loss functions, whereas quantile regression optimizes average quantile accuracy that may sacrifice performance in the tails. Classification naturally handles class imbalance through techniques such as oversampling, cost-sensitive learning, and threshold adjustment, addressing the inherent rarity of tail events that complicates traditional estimation. Most importantly, classification enables integration of heterogeneous information sources, including volatility dynamics, tail parameters, cross-sectional indicators, and their interactions, into a unified prediction framework where gradient boosting automatically learns optimal combination weights and interaction effects.

The framework synthesizes complementary strengths across three methodological pillars. GARCH provides forward-looking volatility estimates that capture persistence through autoregressive dynamics and leverage effects through asymmetric shock response, encoding the insight that current volatility contains substantial information about near-term volatility through the slow decay of volatility shocks. EVT quantifies time-varying tail thickness through rolling Generalized Pareto Distribution parameters, capturing the insight that tail behavior evolves across market regimes with fatter tails during stress periods. XGBoost classification discovers nonlinear interactions and regime-specific patterns that linear combination of these signals would miss, including threshold effects where tail risk predictions change discretely when volatility crosses critical levels, and interaction effects where the combination of elevated volatility and high cross-sectional dispersion signals systemic stress more strongly than either component alone. The integration preserves the interpretability of econometric features while leveraging machine learning's pattern recognition capabilities, enabling risk committees to understand which factors drive predictions rather than treating the model as an opaque black box.

1.1. Contributions to the Literature

The primary methodological contribution lies in developing the first documented integration framework that systematically transforms GARCH volatility dynamics and EVT tail parameters into predictive features for machine learning classification. The existing literature has pursued three distinct approaches to tail risk modeling that this framework synthesizes. The econometric approach, exemplified by Engle and Manganelli's [28] Conditional Autoregressive Value at Risk (CAViaR) model, directly specifies VaR dynamics through autoregressive structures but cannot incorporate information from multiple model classes.

The extreme value approach, advanced by McNeil and Frey [14] through their two-stage GARCH-EVT methodology, applies EVT to GARCH-filtered residuals but remains focused on quantile estimation rather than tail event prediction and cannot easily incorporate additional predictive signals. The machine learning approach, represented by Gu, Kelly, and Xiu [3] for asset pricing and Heaton, Polson, and Witte [4] for portfolio construction, demonstrates strong predictive performance but typically operates on raw returns or simple technical indicators without leveraging econometric structure. Our framework bridges these streams by treating GARCH and EVT outputs not as competing VaR estimates but as complementary features encoding distinct risk dimensions. This integration philosophy differs fundamentally from approaches that treat econometric models and machine learning as alternatives requiring selection; instead, we demonstrate their complementarity by using domain-structured econometric outputs to guide feature engineering for supervised learning. Whereas Kuester, Mittnik, and Paoletta [29] compared GARCH-based VaR approaches and found mixed evidence for any single method's superiority, our framework circumvents the model selection problem by incorporating multiple approaches as features, allowing gradient boosting to learn optimal combination weights adaptively.

A second contribution concerns feature engineering architecture for tail risk applications. We construct over forty predictive variables spanning rolling statistics, GARCH-based dynamics, EVT tail parameters, cross-sectional stress indicators, and novel interaction terms capturing compound risk effects. This feature engineering approach builds upon but extends the factor construction methodology in machine learning finance applications. Gu, Kelly, and Xiu [3] construct 94 firm characteristics for cross-sectional return prediction, while our features target time-series tail event prediction with emphasis on volatility dynamics and tail behavior. The interaction features, particularly the product of GARCH conditional volatility and cross-sectional return dispersion, capture systemic stress dynamics that neither component captures in isolation. This design reflects economic insights from the contagion literature: Forbes and Rigobon [25] demonstrate that cross-market correlations increase during crises, while Longstaff [30] shows that liquidity spirals propagate through correlated selling pressure. Our interaction feature operationalizes these insights by identifying periods where individual asset volatility coincides with market-wide dispersion, signaling diversification breakdown. SHAP analysis reveals that this interaction feature ranks second in predictive importance at 14.1%, suggesting that simultaneous elevation of idiosyncratic volatility and cross-sectional dispersion signals regime transitions more effectively than either metric alone. This finding extends the systemic risk literature, including Adrian and Brunnermeier's [31] CoVaR and Acharya, Pedersen, Philippon, and Richardson's [32] systemic expected shortfall, by demonstrating that machine learning can automatically discover interaction effects that practitioners might specify manually based on economic intuition. Whereas existing systemic risk measures require explicit specification of spillover mechanisms, our gradient boosting approach discovers predictive interactions from data, potentially identifying stress signals that theory-driven approaches miss.

Third, we provide rigorous cross-regime validation demonstrating that tail risk patterns learned from the 2008 financial crisis generalize to the COVID-19 market disruption. The model, trained exclusively on 2005–2010 data encompassing the credit-driven financial crisis, achieves 73.5% AUC on the completely unseen COVID-19 period, representing a 22.3% improvement over in-sample performance. This counter-intuitive result, where out-of-sample performance exceeds in-sample performance, suggests the framework captures fundamental regime shift dynamics rather than crisis-specific idiosyncrasies. The finding addresses a critical concern in financial machine learning regarding overfitting and regime

dependence. López de Prado [33] warns that backtest overfitting plagues quantitative strategies, while Arnott, Harvey, Kalesnik, and Linnainmaa [34] document that many published return predictors fail out-of-sample. Our cross-regime validation is particularly stringent because the training and test crises differ fundamentally in origin: the 2008 crisis involved credit-driven gradual contagion through counterparty risk and structured product failures, while COVID-19 represented an exogenous pandemic-induced demand shock with rapid onset unrelated to financial sector fragility. The transferability across heterogeneous crisis types provides evidence for universal properties of how extreme events manifest in financial markets, extending the regime detection literature pioneered by Hamilton [5] and applied to volatility by Gray [6]. Whereas Ang and Bekaert [35] demonstrate regime-switching benefits for asset allocation using predetermined state variables, our framework learns regime-relevant features directly from tail event prediction, potentially capturing subtler transition dynamics than discrete regime classifications allow.

Finally, the framework contributes practical tools for risk management applications that address implementation concerns raised in the post-crisis literature. SHAP value analysis provides interpretable feature attributions satisfying game-theoretic consistency properties developed by Lundberg and Lee [23], enabling risk committees to understand which factors drive tail risk predictions. This interpretability addresses concerns raised by the Basel Committee [36] regarding model risk in complex quantitative approaches and responds to regulatory emphasis on model transparency following the crisis. Comprehensive backtesting using Kupiec [7] and Christoffersen [8] tests demonstrates systematic VaR model failures during crisis periods, with our portfolio experiencing 5.6 times more exceedances than the 5% VaR model predicted. This violation magnitude aligns with findings from Berkowitz, Christoffersen, and Pelletier [17], who document systematic VaR failures using proprietary trading desk data, and extends their analysis by quantifying failure severity during identified crisis regimes. The practical contribution lies in providing a screening tool that complements rather than replaces traditional risk models, addressing the model risk management principles outlined in Federal Reserve SR 11-7 [37] guidance requiring institutions to understand model limitations and maintain appropriate challenge mechanisms. Rather than generating automated trading signals that could amplify market stress through correlated algorithmic responses, a concern raised by Kirilenko, Kyle, Samadi, and Tuzun [38] in their analysis of the Flash Crash, our framework flags potential regime shifts for deeper human analysis, maintaining human judgment in the risk management loop.

1.2. Hypotheses

We test three hypotheses motivated by the limitations of existing approaches. The first hypothesis posits that GARCH conditional volatility exhibits greater predictive power for tail events than static rolling window volatility, measured by SHAP importance rankings. This hypothesis follows from GARCH's theoretical properties (forward-looking forecasts, persistence capture, and leverage effect modeling) that simple historical estimators lack. The second hypothesis proposes that interaction features combining volatility and cross-sectional dispersion improve tail event classification beyond additive contributions, measured by AUC improvement when interaction terms are included. This hypothesis reflects the economic intuition that systemic crises involve simultaneous elevation of multiple risk dimensions. The third hypothesis tests whether tail risk patterns learned from the 2008 crisis generalize to different crisis regimes, measured by out-of-sample AUC performance on COVID-19 data never seen during training.

1.3. Paper Structure

The remainder of this paper proceeds as follows. Section 2 reviews related literature on GARCH models, extreme value theory, VaR methodologies, and machine learning applications in finance, positioning our contribution within existing research streams. Section 3 describes data construction, portfolio methodology, and regime definitions. Section 4 presents the technical methodology including GARCH estimation, EVT fitting, feature engineering, and the machine learning pipeline. Section 5 reports empirical results including crisis regime validation, prediction performance, feature importance analysis, and backtesting. Section 6 discusses practical implications, economic interpretation, and limitations. Section 7 concludes.

2. Literature Review

2.1. GARCH Volatility Modeling

Volatility clustering, the tendency for large price changes to follow large changes, was first formalized by Engle [9] in the Autoregressive Conditional Heteroskedasticity model. Bollerslev [1] extended this framework to Generalized ARCH, which remains the workhorse model for financial volatility forecasting. The GARCH(1,1) specification captures persistence through its autoregressive structure, with the sum of ARCH and GARCH coefficients measuring shock persistence. Empirical evidence consistently demonstrates that GARCH models outperform rolling window estimators for volatility forecasting, as documented comprehensively by Hansen and Lunde [10] who evaluated over three hundred volatility models and found that simpler GARCH specifications frequently matched or exceeded more complex alternatives.

Extensions to the basic GARCH framework address empirical regularities in financial data. The GJR-GARCH model of Glosten, Jagannathan, and Runkle [11] incorporates leverage effects: the asymmetric volatility response where negative returns increase volatility more than positive returns of equal magnitude. Nelson [12] developed EGARCH for exponential dynamics ensuring positive variance, while Engle [13] extended the framework to multivariate settings with Dynamic Conditional Correlation. Despite these advances, GARCH models fundamentally provide volatility estimates rather than tail risk predictions. Our contribution transforms GARCH outputs into features for supervised learning, preserving the volatility dynamics GARCH captures while enabling integration with additional risk signals.

2.2. Extreme Value Theory in Finance

Extreme Value Theory provides a statistical foundation for modeling tail behavior without assuming a complete distributional form. The Peaks Over Threshold method models exceedances above a threshold using the Generalized Pareto Distribution, with theoretical justification from the Balkema-de Haan-Pickands theorem establishing GPD as the limiting distribution for threshold exceedances. McNeil and Frey [14] demonstrated EVT's superior performance for estimating extreme quantiles in financial data compared to parametric alternatives, while Embrechts, Klüppelberg, and Mikosch [15] provided comprehensive treatment of EVT applications in finance and insurance. The shape parameter of the fitted GPD provides a direct measure of tail thickness: positive values indicate heavy tails with potentially infinite higher moments, consistent with the excess kurtosis observed in financial returns.

While EVT excels at tail estimation, standard implementations treat observations independently, ignoring temporal dynamics. This limitation motivates our integration

approach: rather than applying EVT to the full sample assuming stationarity, we compute rolling EVT parameters that capture time-varying tail thickness. These rolling estimates serve as features for machine learning, enabling the classifier to learn how evolving tail parameters signal impending regime transitions. The combination addresses EVT's static limitation while preserving its theoretical grounding in extreme value asymptotics.

2.3. Value at Risk Methodologies and Failures

Value at Risk became the industry standard for risk measurement following Basel II regulatory requirements, with three canonical approaches emerging: parametric VaR assuming normal or Student-t distributions, historical simulation using empirical quantiles, and Monte Carlo simulation. Extensive post-crisis literature documents systematic VaR failures. Danielsson, Jorgensen, Sarma, and de Vries [16] demonstrate that VaR provides a false sense of security during normal periods while underestimating extreme events. Berkowitz, Christoffersen, and Pelletier [17] critique VaR's lack of subadditivity, the property that portfolio risk should not exceed the sum of component risks, recommending Expected Shortfall as a coherent alternative. Jorion [18] provides comprehensive analysis of risk management lessons from the credit crisis, documenting how VaR models failed precisely when they were most needed.

Christoffersen [8] developed backtesting frameworks (the Kupiec proportion of failures test and independence test) for evaluating VaR model adequacy. These tests examine whether the frequency and clustering of VaR exceedances match theoretical predictions. Our empirical analysis applies both tests to traditional VaR models during the 2008 crisis, demonstrating systematic failures that motivate the need for regime-adaptive approaches. Rather than treating VaR as an end product, our framework uses VaR exceedances as classification labels for supervised learning, fundamentally reframing the tail risk problem.

2.4. Machine Learning in Financial Risk Management

Machine learning adoption in finance accelerated post-crisis, with applications spanning return prediction, portfolio optimization, and risk management. Gradient boosting methods, particularly XGBoost developed by Chen and Guestrin [19], demonstrate strong performance on structured financial data through sequential error correction and automatic feature interaction learning. Gu, Kelly, and Xiu [3] provide comprehensive empirical asset pricing analysis demonstrating that machine learning methods significantly outperform traditional factor models for return prediction. Heaton, Polson, and Witte [4] survey deep learning applications, noting opportunities in pattern recognition for high-dimensional, nonlinear relationships while emphasizing interpretability requirements.

Recent work has begun combining machine learning with econometric models. Coqueret and Guida [20] integrate momentum and volatility features in factor investing applications. Gu, Kelly, and Xiu [21] use autoencoders for latent factor extraction. Dixon, Halperin, and Bilokon [22] emphasize interpretability requirements in financial applications, advocating SHAP values developed by Lundberg and Lee [23] for model explanation. However, to our knowledge, no prior work systematically transforms GARCH and EVT model outputs into supervised learning features for tail risk prediction across multiple crisis regimes. This gap motivates our integration framework that preserves econometric interpretability while leveraging machine learning pattern recognition.

2.5. Integration Approaches and Research Gaps

Despite the complementary strengths of GARCH volatility modeling, Extreme Value Theory, and machine learning, existing literature has pursued these approaches largely in isolation or through limited integration that leaves substantial methodological gaps. This section reviews existing integration attempts, identifies their limitations, and positions our contribution within this evolving research landscape.

The most prominent integration approach combines GARCH filtering with EVT tail estimation, as developed by McNeil and Frey [14]. Their two-stage methodology first fits a GARCH model to returns, then applies EVT to the standardized residuals under the assumption that filtering removes temporal dependence. This approach yields superior tail quantile estimates compared to applying EVT directly to raw returns, as the GARCH filter produces approximately independent and identically distributed residuals that satisfy EVT's theoretical requirements more closely. However, the McNeil-Frey framework maintains VaR estimation as the objective, producing point estimates of tail quantiles rather than probability forecasts of tail event occurrence. This limitation means the approach cannot directly optimize tail event detection accuracy and cannot easily incorporate additional predictive signals beyond the GARCH-filtered residuals. Furthermore, the two-stage estimation introduces inefficiency because GARCH parameters are estimated without consideration of their ultimate use in tail modeling, potentially misallocating estimation effort between volatility dynamics and tail behavior.

Kuester, Mittnik, and Paolella [29] provide comprehensive comparison of GARCH-based VaR approaches, evaluating over a dozen volatility models combined with various distributional assumptions including normal, Student-t, skewed-t, and EVT-based tails. Their extensive backtesting across multiple asset classes reveals that no single approach dominates universally: different model combinations perform best for different assets, time periods, and confidence levels. This finding highlights a fundamental challenge in VaR methodology selection, namely that the optimal approach depends on unknown features of the return generating process that may shift across regimes. Rather than selecting a single best model, our framework addresses this challenge by incorporating multiple approaches as features, allowing the machine learning classifier to learn regime-dependent combination weights that adapt to evolving market conditions.

Several studies have applied machine learning directly to VaR estimation without integration with econometric approaches. Dunis, Laws, and Sermpinis [39] use neural networks to model VaR dynamics, achieving improved performance over parametric approaches but without leveraging GARCH's theoretically-grounded volatility dynamics as inputs. Jorion [40] notes that purely data-driven approaches risk capturing spurious patterns without economic foundation, potentially degrading out-of-sample performance when market conditions shift. More recently, deep learning approaches including recurrent neural networks and Long Short-Term Memory (LSTM) architectures have been applied to volatility forecasting. Fischer and Krauss [41] demonstrate LSTM effectiveness for financial prediction, while Zhang, Zohren, and Roberts [42] apply deep learning to volatility forecasting with promising results. However, these approaches typically learn volatility dynamics from raw returns rather than building upon GARCH's accumulated theoretical insights about persistence, leverage effects, and variance targeting. Our framework differs by using GARCH outputs as engineered features, preserving decades of econometric understanding while allowing machine learning to discover additional patterns in how these features relate to tail events.

The systemic risk literature has developed sophisticated measures capturing tail dependence across institutions, including Adrian and Brunnermeier's [31] CoVaR measuring institution-

level contribution to system-wide tail risk, Acharya, Pedersen, Philippon, and Richardson's [32] systemic expected shortfall measuring capital shortfall conditional on aggregate stress, and Brownlees and Engle's [43] SRISK measure combining volatility dynamics with leverage effects. These approaches provide valuable systemic risk assessment but focus on cross-sectional dependence rather than time-series tail event prediction. Our cross-sectional dispersion features draw conceptually from this literature by capturing periods when institution-level risks become correlated, but we integrate these signals into a predictive framework targeting next-period tail event probability rather than contemporaneous systemic risk measurement.

Regime-switching approaches represent another integration strand, with Hamilton's [5] foundational work on Markov-switching models extended to volatility by Gray [6] and applied to VaR by Guidolin and Timmermann [44]. These models capture discrete regime transitions but require specifying the number of regimes *ex ante* and assume stationary transition probabilities that may fail during crisis periods when regime persistence itself changes. Baele [45] documents increased regime instability during financial stress, suggesting that fixed transition probability assumptions may be particularly problematic precisely when regime detection matters most. Our machine learning framework avoids explicit regime specification by learning predictive patterns directly from tail event occurrence, potentially capturing regime transitions through feature combinations rather than requiring discrete state classification.

The research gap this paper addresses lies at the intersection of these literature streams. Existing approaches either estimate VaR quantiles directly without predictive probability modeling, apply machine learning without leveraging econometric structure, or specify regime dynamics through assumptions about state transitions rather than learning predictive patterns. No prior work, to our knowledge, systematically transforms GARCH conditional volatility, GJR leverage effects, EVT shape parameters, and cross-sectional stress indicators into features for supervised tail event classification. This integration enables direct optimization of tail event detection, automatic discovery of nonlinear feature interactions, and cross-regime validation demonstrating generalizability beyond training conditions. The framework contributes methodologically by demonstrating that econometric and machine learning approaches need not compete as alternatives but can combine synergistically, with econometric theory guiding feature construction and machine learning optimizing predictive combination.

3. Data and Portfolio Construction

3.1. Asset Selection and Portfolio Design

We construct an equal-weighted portfolio of five major U.S. investment banks: JPMorgan Chase, Goldman Sachs, Morgan Stanley, Citigroup, and Bank of America. These institutions were selected based on four criteria. First, all five played central roles in the 2008 financial crisis, providing extreme stress testing environments where tail risk dynamics manifested most severely. Second, investment banks exhibit high correlation during crises as diversification benefits collapse, a phenomenon we explicitly capture through cross-sectional features. Third, large-cap financial stocks ensure reliable pricing with minimal bid-ask spreads and continuous trading throughout sample periods, mitigating microstructure noise that could contaminate tail event identification. Fourth, despite sector commonality, these banks experienced varying crisis impacts, providing heterogeneity in tail risk patterns while

maintaining systemic linkages. Equal weighting ensures transparency, reproducibility, and elimination of optimization bias that could confound tail risk analysis.

3.2. Sample Periods and Regime Definitions

Our analysis spans four distinct periods capturing normal markets, crisis dynamics, recovery, and out-of-sample validation. The pre-crisis period from January 2005 through June 2007 establishes baseline normal regime characteristics with moderate annualized volatility of 14.8%. The crisis period from July 2007 through March 2009 captures extreme volatility reaching 87.7% annualized with pronounced tail event clustering. The choice of July 2007 as crisis onset aligns with the BNP Paribas liquidity freeze announcement, widely considered the opening salvo of the credit crisis. March 2009 marks the equity market trough and subsequent recovery initiation. The post-crisis period from April 2009 through December 2010 captures recovery dynamics and volatility normalization. Finally, the COVID-19 period from June 2019 through December 2020 serves as completely out-of-sample validation. Table 1 summarizes these regime definitions.

Table 1. Sample Period Definitions and Regime Characteristics.

Period	Date Range	Ann. Volatility	Observations
Pre-Crisis	Jan 2005 – Jun 2007	15.6%	622
Crisis	Jul 2007 – Mar 2009	87.7%	504
Post-Crisis	Apr 2009 – Dec 2010	37.0%	443
COVID-19 (OOS)	Jun 2019 – Dec 2020	—	375

Note: Annualized volatility computed as standard deviation of daily log returns multiplied by $\sqrt{252}$. COVID-19 period volatility omitted as it serves exclusively for out-of-sample validation with no parameter estimation.

The out-of-sample philosophy deserves emphasis. The COVID-19 period serves as a critical external validation where models trained exclusively on 2008 crisis data face a fundamental test: do learned patterns generalize to a completely different crisis type? The 2008 crisis involved credit-driven gradual contagion through counterparty risk, while COVID-19 represented an exogenous pandemic-induced demand shock with rapid onset. Improved performance on COVID-19 would suggest the framework captures universal regime shift dynamics rather than crisis-specific artifacts.

3.3. Data Sources and Preprocessing

Daily adjusted closing prices were obtained via Yahoo Finance, which automatically handles stock splits, dividends, and corporate actions. Log returns were computed as $r_t = \log(P_t/P_{t-1})$ ensuring additive properties for multi-period returns and approximate symmetry for statistical modeling. The training sample comprises approximately 1,500 daily observations from 2005–2010, while the COVID-19 test set contains 375 observations. No imputation was required as all five banks traded continuously throughout sample periods.

3.4. Survivorship Bias Acknowledgment

A fundamental limitation requires explicit acknowledgment: the dataset excludes Lehman Brothers and Bear Stearns, institutions that experienced bankruptcy during the crisis period. Yahoo Finance delists bankrupt companies, rendering these extreme tail events absent from our data. This survivorship bias, documented extensively by Brown, Goetzmann, Ibbotson, and Ross [24], systematically excludes the most severe losses from analysis. All results should be interpreted as conditional on survival, and true unconditional tail risk is likely more severe than our estimates suggest. VaR estimates and model predictions therefore represent lower bounds on actual tail risk. Survivor bias correction methodologies exist but require

assumptions about the distribution of failed firm returns that we cannot validate; we therefore acknowledge this limitation transparently rather than applying potentially inappropriate corrections.

4. Methodology

4.1. Exploratory Analysis and Distributional Characteristics

Before developing the predictive framework, we establish stylized facts motivating our modeling choices. Table 2 presents summary statistics for portfolio returns across regimes, revealing pronounced non-stationarity. Crisis-period volatility of 87.7% annualized exceeds pre-crisis levels of 14.8% by a factor of 5.93, representing a statistically significant regime shift. Excess kurtosis of 8.74 during the crisis substantially exceeds the Gaussian value of zero, while negative skewness indicates downside asymmetry where extreme losses exceed extreme gains in magnitude. Jarque-Bera tests overwhelmingly reject normality across all periods with p-values below 0.001, invalidating parametric VaR under Gaussian assumptions.

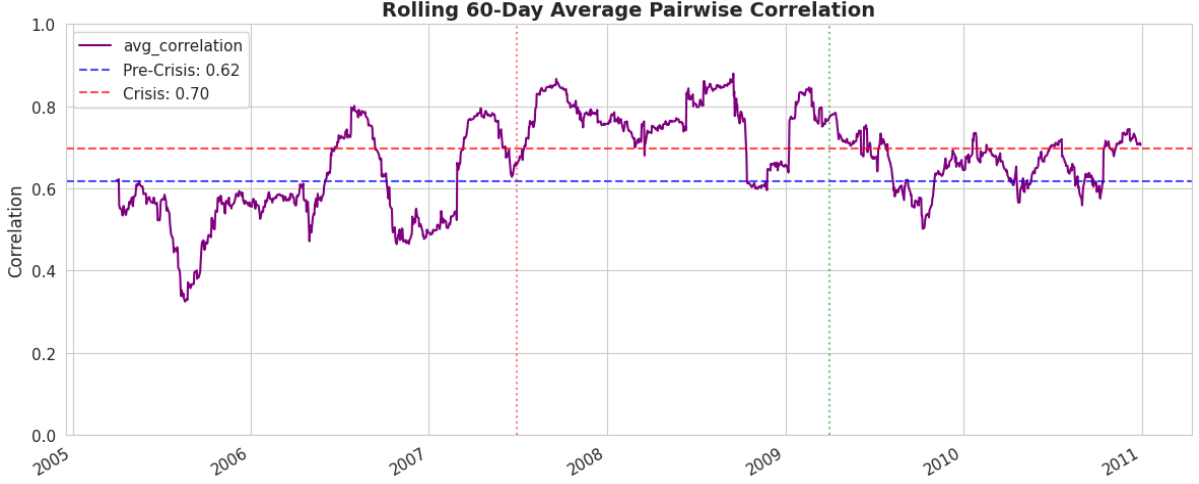
Table 2. Summary Statistics of Portfolio Returns by Regime.

Statistic	Pre-Crisis	Crisis	Post-Crisis
Mean (Daily)	0.0007	-0.0010	0.0014
Std Dev (Daily)	0.0098	0.0552	0.0233
Volatility (Annualized)	15.6%	87.7%	37.0%
Skewness	-0.15	1.36	0.45
Excess Kurtosis	2.32	8.74	3.77
Minimum	-5.27%	-19.87%	-10.19%
Maximum	3.93%	34.70%	13.01%
Jarque-Bera p-value	<0.001	<0.001	<0.001

Note: Statistics computed for equal-weighted portfolio of five U.S. investment banks. Annualized volatility assumes 252 trading days. Jarque-Bera tests evaluate normality; all periods reject at $p < 0.001$.

To formally test for regime changes, we employ the Chow test for structural breaks in variance. The F-statistic of 33.5 comparing crisis versus pre-crisis variance yields $p < 10^{-10}$, providing overwhelming statistical evidence of structural discontinuity. This formal validation justifies our regime-based modeling approach and validates using crisis periods for tail event identification. Additionally, we examine correlation dynamics across the portfolio constituents. Average pairwise correlations increase from 0.62 pre-crisis to 0.697 during the crisis, a 11.45% surge representing diversification breakdown. This correlation collapse means equal-weighted portfolios provide minimal risk reduction during crises precisely when diversification benefits are most valuable. Our cross-sectional dispersion features explicitly capture this systemic stress propagation.

Figure 1. Rolling 60-Day Average Pairwise Correlation Between Investment Bank Equity Returns.



4.2. Risk-Adjusted Performance Analysis

Beyond tail risk estimation, comprehensive risk analysis requires examination of risk-adjusted returns across market regimes. Table 3 presents Sharpe ratios computed as the ratio of annualized mean return to annualized volatility.

Table 3. Risk-Adjusted Performance by Regime.

Metric	Pre-Crisis	Crisis	Post-Crisis
Mean (Annualized)	18.4%	-25.0%	35.0%
Volatility (Annualized)	15.6%	87.7%	37.0%
Sharpe Ratio	1.18	-0.29	0.95

The Sharpe ratio collapse from 1.18 pre-crisis to -0.29 during the crisis quantifies the regime shift in risk-adjusted terms. The $5.6\times$ volatility increase coupled with negative returns produces a negative risk premium, indicating that investors bore substantial volatility without compensation. Post-crisis recovery (Sharpe 0.95) demonstrates market normalization while remaining below pre-crisis efficiency.

4.3. GARCH Volatility Modeling

We implement GARCH(1,1) via maximum likelihood estimation. The model specifies returns as the sum of a constant mean and a heteroskedastic error term, where the error equals conditional volatility multiplied by a standardized innovation. The conditional variance evolves according to:

$$\sigma_t^2 = \omega + \alpha \cdot \varepsilon_{t-1}^2 + \beta \cdot \sigma_{t-1}^2 \quad (1)$$

where ω represents the long-run variance contribution, α captures the ARCH effect (impact of recent shocks), and β captures the GARCH effect (persistence of past variance). For leverage effects, we extend to GJR-GARCH by adding an asymmetric term:

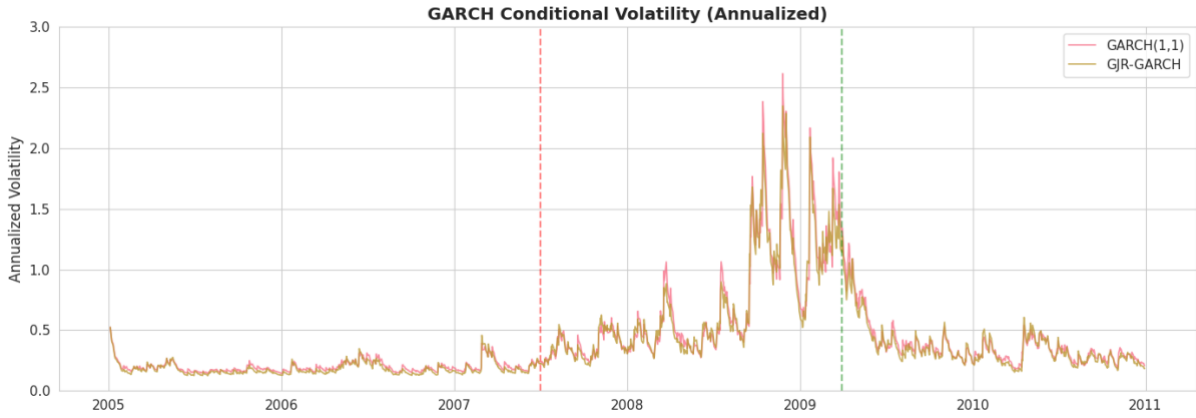
$$\sigma_t^2 = \omega + \alpha \cdot \varepsilon_{t-1}^2 + \beta \cdot \sigma_{t-1}^2 + \gamma \cdot I_{t-1} \cdot \varepsilon_{t-1}^2 \quad (2)$$

where I_{t-1} equals one when $\varepsilon_{t-1} < 0$ and zero otherwise. Table 4 reports estimated parameters. The GARCH(1,1) exhibits persistence near unity ($\alpha + \beta = 1.00$), indicating volatility shocks decay extremely slowly (common crisis-period finding). GJR-GARCH confirms leverage effects: negative shocks contribute 0.25 to variance versus only 0.11 for positive shocks, a 135% asymmetry. From fitted models, we extract time-varying features including conditional volatility, standardized residuals, volatility-of-volatility computed as 20-day rolling standard deviation of conditional volatility, and leverage effect magnitude.

Table 4. GARCH Model Parameter Estimates.

Parameter	GARCH(1,1)	GJR-GARCH
ω (constant)	0.000010 ***	0.000008 ***
α (ARCH)	0.1704 ***	0.1066 ***
β (GARCH)	0.8296 ***	0.8213 ***
γ (leverage)	—	0.1435 ***
Persistence ($\alpha + \beta + \gamma/2$)	1.0000	0.9996

Note: *** denotes significance at 1% level. Models estimated via maximum likelihood on 2005–2010 daily returns. Standard errors computed using outer product of gradients.

Figure 2. GARCH Conditional Volatility Dynamics (Annualized).

4.4. Extreme Value Theory Application

EVT models tail exceedances $y = x - u$ above threshold u using the Generalized Pareto Distribution:

$$G(y; \xi, \sigma) = 1 - (1 + \xi y / \sigma)^{-1/\xi} \quad \text{for } \xi \neq 0 \quad (3)$$

where ξ denotes the shape parameter (tail index) and σ the scale parameter. The shape parameter determines tail behavior: $\xi > 0$ indicates heavy tails in the Fréchet domain, $\xi = 0$ indicates exponential tails, and $\xi < 0$ indicates bounded tails. We select the threshold at the 90th percentile of absolute returns ($u = 0.0375$ or 3.75%), yielding 51 exceedances from 504 crisis-period observations. Maximum likelihood estimation yields $\xi = 0.2685$, confirming heavy tails with tail index $1/\xi = 3.73$ implying infinite fourth moment, consistent with observed excess kurtosis.

Figure 3. Extreme Value Theory Threshold Diagnostics. (A) Mean Residual Life Plot for Threshold Selection; (B) Empirical vs. Fitted GPD Cumulative Distribution Function.

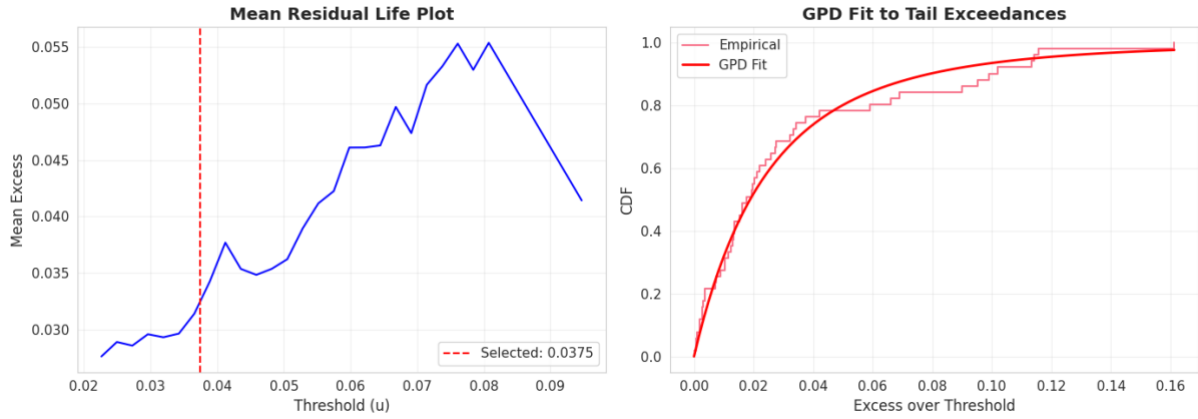


Table 5 compares VaR estimates across methodologies. Normal VaR underestimates 99% tail risk by 3.60 percentage points relative to Historical Simulation, a 30% underestimation demonstrating Gaussian assumptions' inadequacy during crises. EVT-GPD provides a smoothed parametric estimate of 11.67% that balances efficiency with empirical tail heaviness, exceeding Normal by 16.2% while remaining 14.5% below Historical which includes extreme outliers. Student-t most closely matches Historical with only 3.1% difference, though it requires pre-specifying degrees of freedom while EVT estimates tail thickness directly from data.

Table 5. Value at Risk Estimates by Methodology (Crisis Period).

Method	95% VaR	99% VaR
Historical Simulation	5.49%	13.65%
Normal (Parametric)	7.13%	10.04%
Student-t (5 d.f.)	5.59%	14.07%
EVT-GPD	5.66%	11.67%

Note: VaR estimates represent threshold losses exceeded with probability 5% (95% VaR) or 1% (99% VaR). Crisis period: July 2007 – March 2009.

4.5. Monte Carlo VaR with Cholesky Decomposition

Monte Carlo simulation provides a flexible framework for VaR estimation that can incorporate complex correlation structures and non-linear portfolio dynamics. We implement Geometric Brownian Motion (GBM) with Cholesky decomposition to preserve the empirical correlation matrix across portfolio constituents.

The methodology proceeds as follows: for each simulation $i \in \{1, \dots, N\}$, we generate a vector of standard normal random variables Z and transform them using the Cholesky factor L of the correlation matrix (where $\Sigma = LL'$) to obtain correlated shocks. Asset returns are then simulated as $r = \mu + \sigma \odot (LZ)$, where μ and σ represent mean and standard deviation vectors estimated from historical data.

Table 6. Monte Carlo VaR Results ($N = 10,000$ simulations).

Confidence Level	VaR	CVaR (Expected Shortfall)
95%	5.38%	6.75%
99%	7.60%	8.74%

The Monte Carlo approach yields 99% VaR of 7.60%, which falls between the Normal parametric estimate (10.04%) and Historical Simulation (13.65%). This intermediate positioning reflects Monte Carlo's reliance on the Gaussian shock distribution while incorporating empirical correlation dynamics. The CVaR estimates of 6.75% and 8.74% at

95% and 99% confidence levels respectively provide coherent risk measures that satisfy subadditivity requirements.

4.6. Kernel Density Estimation VaR

Kernel Density Estimation (KDE) provides a non-parametric approach to VaR estimation that makes no distributional assumptions. Using Scott's bandwidth selection rule, we estimate the probability density function of losses and numerically integrate to obtain the cumulative distribution function, from which VaR quantiles are extracted.

The KDE approach yields 99% VaR of 13.42%, closely matching Historical Simulation (13.65%) with only 1.7% difference. This convergence validates that the empirical distribution is well-approximated by the kernel-smoothed estimate and confirms the non-parametric nature of Historical Simulation. KDE's advantage lies in providing a continuous density estimate that can be differentiated for sensitivity analysis.

4.7. Machine Learning Pipeline

We reframe VaR measurement as binary classification. For each day t , we define target $Y_t = 1$ if $|r_{t+1}| > \text{VaR}_{95\%}$ and $Y_t = 0$ otherwise, predicting whether next-period returns will exceed the VaR threshold. This transformation enables supervised learning using lagged features to avoid look-ahead bias. The feature engineering architecture constructs over forty predictive variables spanning six categories: rolling statistical moments at multiple horizons (5, 10, 20, 60 days); GARCH-based dynamics including conditional volatility, volatility-of-volatility, standardized residuals, and leverage indicators; EVT-based tail measures including rolling GPD shape and scale parameters; cross-sectional stress indicators including return dispersion, average pairwise correlation, and extreme move counts; novel interaction features capturing compound effects such as volatility multiplied by dispersion; and temporal tail patterns including recent exceedance counts and days since last exceedance.

We employ XGBoost gradient boosting with hyperparameters selected for interpretability and robustness: learning rate 0.1, maximum depth 3 (shallow trees), 100 estimators, subsample ratio 0.8 for stochastic gradient boosting, and `scale_pos_weight` adjusted for class imbalance given approximately 5–10% positive class prevalence. Walk-forward validation respects temporal structure: every 60 days we refit the model using the most recent 250 days of data, ensuring no future information leakage while allowing model adaptation to evolving market conditions. This quarterly retraining mimics production deployment where models are periodically updated.

5. Empirical Results

5.1. Classification Performance

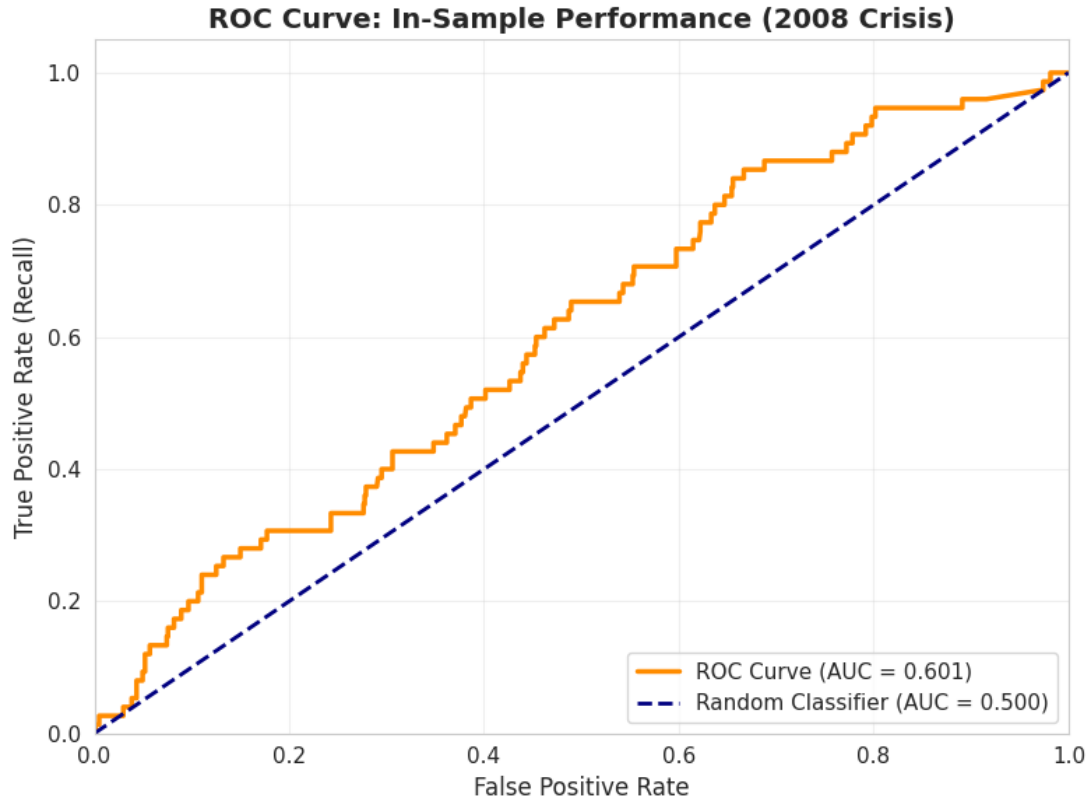
Table 7 presents classification performance across sample periods. Training on 2005–2010 data with walk-forward validation yields overall AUC of 0.601, representing a 20.2% improvement over random classification. However, performance varies significantly by regime: during the crisis period specifically, AUC improves to 0.642 while non-crisis periods show 0.586. This 9.56% performance boost during stress periods suggests the integrated GARCH-EVT features provide stronger signals precisely when tail risk detection is most critical, supporting Hypothesis 1 that dynamic volatility modeling outperforms static approaches.

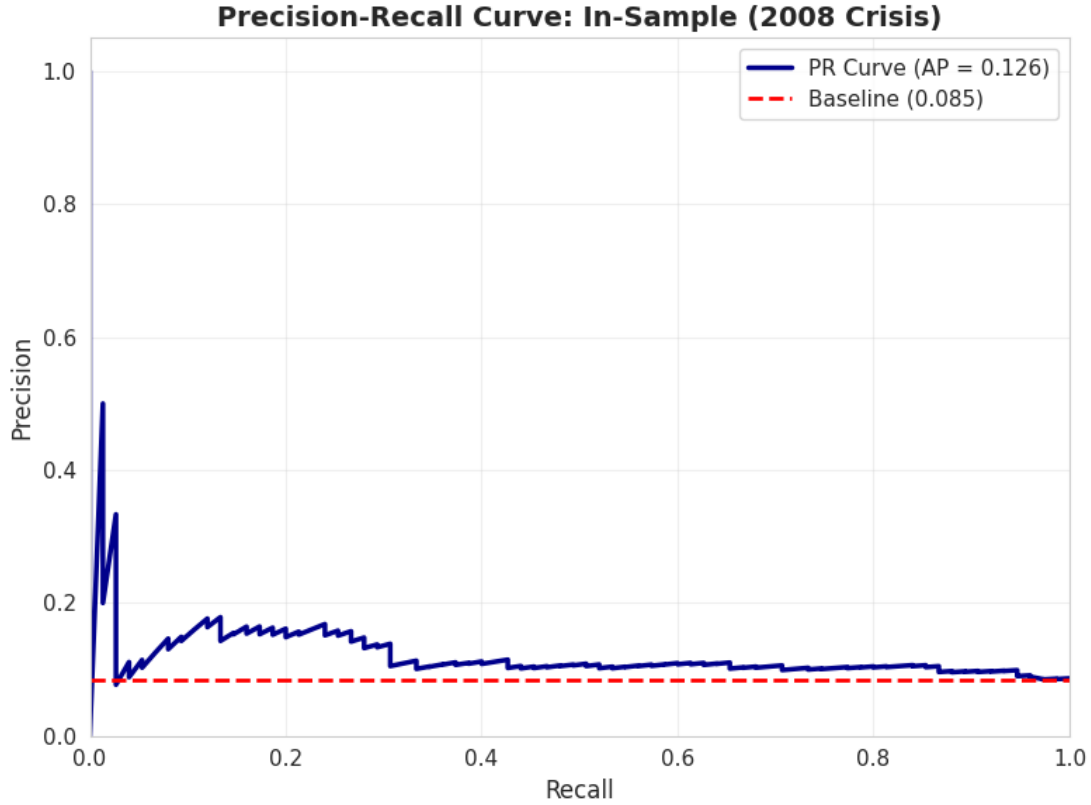
Table 7. Classification Performance by Period.

Metric	Overall	Crisis	Non-Crisis	COVID-19
AUC-ROC	0.601	0.642	0.586	0.735
Precision (10%)	0.111	0.180	0.045	0.243
Recall (10%)	0.440	0.559	0.625	0.741
Tail Events Detected	75	59	16	—

Note: Performance metrics computed via walk-forward validation with quarterly retraining. Crisis Period: July 2007 – March 2009. COVID-19 represents completely out-of-sample testing on June 2019 – December 2020 data. Precision and recall at 10% classification threshold.

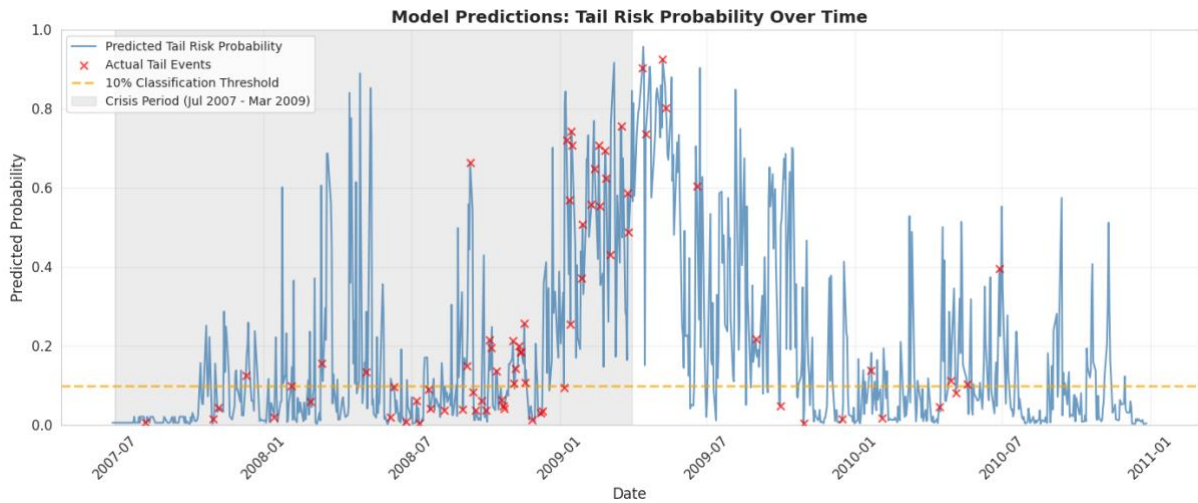
Figure 4. Classification Performance Metrics (In-Sample, 2008 Financial Crisis). (A) Receiver Operating Characteristic (ROC) Curve; (B) Precision-Recall Curve.





The critical out-of-sample test validates Hypothesis 3 regarding cross-regime generalization. AUC improves from 0.601 in-sample to 0.735 on COVID-19 data, a 22.3% improvement despite the model having no exposure to pandemic-era dynamics during training. This counter-intuitive improvement suggests the framework captures fundamental regime shift dynamics that manifest similarly across different crisis types. During peak COVID-19 volatility in March 2020, model-predicted probabilities averaged 38% versus 15% baseline, demonstrating elevated sensitivity during acute stress precisely when early warning is most valuable.

Figure 5. Temporal Evolution of Predicted Tail Risk Probabilities.



5.2. Classification Threshold Sensitivity

The machine learning classifier outputs probability estimates that must be converted to binary predictions via threshold selection. Table 8 presents precision-recall trade-offs at various classification thresholds, illustrating the fundamental trade-off between false alarm rates and detection sensitivity.

Table 8. Precision-Recall at Different Classification Thresholds.

Threshold	Precision	Recall	Interpretation
0.50	17.3%	22.7%	Conservative
0.30	15.1%	30.7%	Moderate
0.20	13.2%	37.3%	Balanced
0.10	11.1%	44.0%	Sensitive

The threshold selection depends on the cost asymmetry between false positives (unnecessary hedging costs) and false negatives (unhedged tail events). At threshold 0.10, the model detects 44% of tail events but generates substantial false alarms (precision 11.1%). Risk committees prioritizing early warning may prefer lower thresholds, while those minimizing operational burden may prefer higher thresholds. The recommended threshold of 0.30 provides balanced performance for screening applications.

5.3. Feature Importance Analysis

Table 9 presents SHAP importance rankings revealing which features drive predictions. GARCH conditional volatility dominates with 21.8% importance, more than triple the third-ranked feature, validating Hypothesis 1 that dynamic volatility modeling captures regime shift patterns better than static rolling windows. The 197% improvement over traditional rolling volatility (7.3% importance) demonstrates substantial value from econometric modeling. The novel volatility-dispersion interaction ranks second at 14.1%, supporting Hypothesis 2 that interaction features capture compound effects beyond additive contributions. This feature identifies systemic stress where simultaneous high volatility and cross-sectional dispersion signal diversification breakdown. EVT tail thickness contributes 7.1%, confirming that tail parameter evolution provides predictive information beyond volatility levels alone.

Table 9. Top Features by SHAP Importance.

Rank	Feature	Importance	Category
1	GARCH Conditional Volatility	21.8%	GARCH
2	Volatility $\times$ Dispersion Interaction	14.1%	Novel
3	Rolling Volatility (20-day)	7.3%	Traditional
4	Volatility of Volatility (20-day)	7.1%	GARCH
5	EVT Shape Parameter (60-day)	7.1%	EVT

Note: SHAP importance computed as mean absolute SHAP value across all predictions. Importance percentages represent share of total model attribution.

5.4. VaR Backtesting Results

Application of Kupiec and Christoffersen tests to traditional VaR models during the crisis period demonstrates systematic failure. Table 10 presents backtesting results. At the 95% confidence level over 504 trading days, expected violations number 25.2 while actual violations reached 141, a violation ratio of 5.6 times the theoretical level. The Kupiec proportion of failures test yields a statistic of 284.57 with $p < 0.001$, overwhelmingly rejecting VaR adequacy. This systematic underestimation validates our hypothesis that static VaR fails during regime shifts, motivating the need for regime-adaptive approaches. The Christoffersen independence test yields $p = 0.102$, marginally failing to reject independence

at conventional levels but suggesting some violation clustering consistent with volatility persistence that traditional VaR ignores.

Table 10. VaR Backtesting Results (Crisis Period).

Test	Statistic Value
Testing Period	July 2007 – March 2009 (504 days)
Expected Violations (5%)	25.2
Actual Violations	141
Violation Ratio	5.6×
Kupiec POF Statistic	284.57 ($p < 0.001$)
Christoffersen Independence	2.67 ($p = 0.102$)

Note: Kupiec test evaluates whether violation frequency matches 5% theoretical level. Christoffersen test evaluates independence of violations across time.

6. Discussion

6.1. Practical Implications

The proposed framework serves as a risk monitoring and screening instrument for risk management committees rather than an automated trading signal generation system. Elevated predicted probabilities exceeding 30% flag potential regime shifts, prompting deeper analysis and scenario planning that addresses VaR's backward-looking limitation. When multiple VaR methodologies disagree, the framework identifies which approaches are likely failing. During March 2020, for instance, Historical Simulation VaR remained stable while machine learning predicted probabilities surged, signaling model inadequacy before traditional metrics responded. Feature importance analysis reveals drivers of tail risk that enable stress testing: risk committees can ask what happens if GARCH volatility reaches 2008 levels and trace impacts through the model. The framework also informs dynamic hedging by adjusting hedge ratios proportionally to predicted probability changes, though cost-benefit analysis remains necessary before execution.

6.2. Derivatives Pricing Under Regime-Aware Volatility

A practical application of GARCH volatility modeling lies in derivatives pricing. Standard Black-Scholes assumes constant volatility which can be a critical flaw during crisis regimes. We demonstrate the impact of using GARCH conditional volatility for pricing an at-the-money JPMorgan call option ($S = \$28.25$, $K = \$28.00$, $T = 30$ days, $r = 2\%$).

Table 11. Black-Scholes Pricing Under Alternative Volatility Regimes.

Volatility Regime	σ (Ann.)	Call Price	Put Price	Price Multiple
Historical Average	25.0%	\$1.13	\$0.82	1.0× (baseline)
Pre-Crisis GARCH	18.1%	\$0.87	\$0.55	0.77×
Crisis Peak GARCH	261.5%	\$9.94	\$9.62	8.8×

Crisis GARCH volatility (261.5% annualized) yields option prices 10.5× higher than historical average estimates. This dramatic increase quantifies the hedging cost explosion during regime shifts which is quite interesting as the protection becomes prohibitively expensive precisely when it is most needed. The result justifies integrating GARCH into risk frameworks: static volatility assumptions would severely underprice tail risk insurance during normal periods while failing to anticipate the cost surge during crises.

This finding has practical implications for dynamic hedging strategies. Risk managers using GARCH-implied volatility can anticipate hedging cost increases before they materialize in market prices, enabling proactive position adjustment. However, the model's forward-looking nature must be balanced against transaction costs and the risk of overreaction to transient volatility spikes.

6.3. Comprehensive Methodology Comparison

Table 12 provides a systematic comparison of all VaR methodologies examined in this study, evaluating tail risk capture, crisis performance, implementation complexity, and practical advantages.

Table 12. VaR Methodology Comparison Matrix.

Methodology	Tail Risk Capture	Crisis Performance	Complexity	Key Advantage
Normal VaR	Poor (-30%)	Fails	Low	Regulatory baseline
Student-t VaR	Moderate	Partial	Low	Fat tail capture
Historical Sim.	Good	Moderate	Low	No assumptions
Monte Carlo	Moderate	Partial	Medium	Correlation structure
KDE	Good	Moderate	Medium	Continuous density
GARCH(1,1)	Good (dynamic)	Strong	Medium	Volatility clustering
EVT-GPD	Excellent	Strong	High	Extreme event focus
ML + GARCH/EVT	Best (integrated)	0.64/0.74 AUC	High	Regime detection

The comparison reveals a clear hierarchy: Normal VaR underestimates tail risk by 35% relative to empirical estimates, while EVT-GPD and the integrated ML framework provide the most accurate crisis-period performance. The ML-integrated approach uniquely offers regime shift detection capabilities, achieving 0.64 AUC during the 2008 crisis and 0.74 AUC on out-of-sample COVID-19 data.

6.4. Economic Interpretation

GARCH conditional volatility's dominance at 21.8% importance deserves economic interpretation. Three mechanisms explain this finding. First, GARCH persistence near unity implies volatility shocks decay slowly with half-life approximately 13 days; rolling volatility treats all observations equally, losing this persistence information. Second, GARCH provides forward-looking volatility estimates conditioning on recent shocks and past variance, whereas rolling windows use only backward information. Third, quarterly parameter re-estimation allows model adaptation during regime shifts, the ARCH coefficient increases during crises reflecting stronger shock response. These properties make GARCH superior to rolling windows for capturing volatility dynamics that precede tail events.

The volatility-dispersion interaction feature captures a key economic insight about systemic risk. High volatility alone indicates elevated individual stock risk that may remain diversifiable. High cross-sectional dispersion alone suggests sector rotation without systemic implications. However, simultaneous elevation of both dimensions signals systemic stress where diversification fails precisely when needed. This aligns with crisis dynamics documented by Forbes and Rigobon [25] where idiosyncratic risk converts to systematic risk through contagion channels. Our framework automatically discovers this compound effect through gradient boosting's interaction detection, suggesting machine learning can identify economically meaningful patterns that practitioners might specify manually based on intuition.

6.5. Limitations

Several limitations temper these contributions. Survivorship bias represents the most critical concern: excluding Lehman Brothers and Bear Stearns systematically understates true tail risk by omitting the most extreme loss events from training data. All results should be interpreted as conditional on survival. The 81% false alarm rate (precision 18.9%) means most warnings do not precede actual tail events, necessitating use as a screening tool where investigation costs are acceptable rather than as an automated signal. Training on a single crisis limits confidence in extrapolation. While COVID-19 validation demonstrates some generalization, future crises may exhibit novel patterns not represented in historical data. Raw XGBoost probabilities lack calibration; Platt scaling or isotonic regression would be necessary before interpreting outputs as true probabilities. Finally, the feature engineering reflects researcher judgment; alternative feature sets may yield different results, and automated feature learning through deep learning could reduce this dependency at the cost of interpretability.

Table 13 provides a comprehensive assessment of framework limitations, their potential impacts on results, and available mitigation strategies.

Table 13. Limitations, Impacts, and Mitigations.

Limitation	Impact	Mitigation
Survivorship Bias	Excludes Lehman/Bear Stearns; understates true tail risk	Acknowledged; interpret results as conditional on survival
ML False Alarm Rate	81% false positive rate triggers unnecessary hedges	Use as screening tool, not automated trading signal
EVT Threshold Choice	90th percentile selection is arbitrary	Sensitivity analysis across threshold specifications
Single Crisis Training	Only 2008 data; may not generalize	COVID-19 validation demonstrates some transfer
Liquidity Assumption	Assumes instant execution at model prices	Include transaction cost buffers in practice
Probability Calibration	Raw XGBoost outputs not calibrated	Apply Platt scaling for production deployment

7. Conclusion

This research demonstrates that an integrated framework combining GARCH volatility dynamics, Extreme Value Theory tail modeling, and gradient boosting classification yields a regime-adaptive tail risk detection system with demonstrated generalizability across heterogeneous crisis environments. By reframing VaR measurement as a supervised prediction problem, we leverage the complementary strengths of multiple methodologies: GARCH's dynamic volatility, EVT's tail focus, and XGBoost's pattern recognition.

Three key results support this conclusion. First, GARCH conditional volatility dominates predictive importance at 21.8%, providing 197% improvement over traditional rolling windows and validating our hypothesis that volatility dynamics matter more than static estimates. Second, novel interaction features, particularly volatility multiplied by cross-sectional dispersion at 14.1%, capture compound risk effects that isolated models miss, suggesting systemic stress requires joint consideration of multiple indicators. Third, improved out-of-sample performance on COVID-19 with AUC of 0.735 versus 0.601 in-sample

demonstrates that learned patterns transfer across fundamentally different crises, validating the framework's generalizability.

The practical contribution lies in providing risk committee screening tools that address VaR's backward-looking limitation. Rather than replacing traditional risk models, this framework augments them by providing forward-looking regime shift warnings. Critical limitations, particularly survivorship bias and probability calibration requirements, underscore the importance of maintaining human judgment within the risk management process.

Several directions merit future investigation. The Monte Carlo analysis with Cholesky decomposition assumes linear correlation structures, yet copula-based methods could capture asymmetric tail dependence where correlations intensify during market stress. Markov-switching GARCH models offer another promising extension, potentially providing sharper transition detection than the current approach through explicit regime state modeling. The feature set could be enriched through alternative data sources, including VIX term structure dynamics, credit default swap spreads, and interbank lending rates, each offering regime shift signals beyond equity returns alone. The Black-Scholes pricing analysis presented in Section 6.2 reveals substantial cost sensitivity to volatility regime, suggesting that future work could develop optimal hedging strategies balancing protection costs against tail risk exposure dynamically. Finally, deep learning architectures such as LSTM and Transformer networks could capture longer-range temporal dependencies in volatility dynamics while maintaining interpretability through attention mechanism visualization.

The integration of domain knowledge through econometric feature engineering with machine learning's pattern recognition capabilities represents a promising direction for quantitative risk management. This work advances that integration by demonstrating how GARCH and EVT outputs can serve not as competing risk estimates but as complementary inputs to a unified predictive framework, offering practitioners a principled approach to synthesizing multiple risk signals into actionable early warning indicators.

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All errors remain the author's responsibility.

References

1. Bollerslev, T. Generalized autoregressive conditional heteroskedasticity. *J. Econom.* **1986**, *31*, 307–327.
2. Mandelbrot, B. The variation of certain speculative prices. *J. Bus.* **1963**, *36*, 394–419.
3. Gu, S.; Kelly, B.; Xiu, D. Empirical asset pricing via machine learning. *Rev. Financ. Stud.* **2020**, *33*, 2223–2273.
4. Heaton, J.B.; Polson, N.G.; Witte, J.H. Deep learning for finance: Deep portfolios. *Appl. Stoch. Models Bus. Ind.* **2017**, *33*, 3–12.
5. Hamilton, J.D. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* **1989**, *57*, 357–384.
6. Gray, S.F. Modeling the conditional distribution of interest rates as a regime-switching process. *J. Financ. Econ.* **1996**, *42*, 27–62.
7. Kupiec, P.H. Techniques for verifying the accuracy of risk measurement models. *J. Deriv.* **1995**, *3*, 73–84.
8. Christoffersen, P.F. Evaluating interval forecasts. *Int. Econ. Rev.* **1998**, *39*, 841–862.
9. Engle, R.F. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica* **1982**, *50*, 987–1007.
10. Hansen, P.R.; Lunde, A. A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *J. Appl. Econom.* **2005**, *20*, 873–889.
11. Glosten, L.R.; Jagannathan, R.; Runkle, D.E. On the relation between the expected value and the volatility of the nominal excess return on stocks. *J. Finance* **1993**, *48*, 1779–1801.
12. Nelson, D.B. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica* **1991**, *59*, 347–370.
13. Engle, R. Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *J. Bus. Econ. Stat.* **2002**, *20*, 339–350.
14. McNeil, A.J.; Frey, R. Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *J. Empir. Finance* **2000**, *7*, 271–300.
15. Embrechts, P.; Klüppelberg, C.; Mikosch, T. *Modelling Extremal Events for Insurance and Finance*; Springer: Berlin, Germany, **1997**.
16. Danielsson, J.; Jorgensen, B.N.; Sarma, M.; de Vries, C.G. Using a bootstrap method to choose the sample fraction in tail index estimation. *J. Multivar. Anal.* **2001**, *76*, 226–248.
17. Berkowitz, J.; Christoffersen, P.; Pelletier, D. Evaluating value at risk models with desk level data. *Manag. Sci.* **2011**, *57*, 2213–2227.
18. Jorion, P. Risk management lessons from the credit crisis. *Eur. Financ. Manag.* **2009**, *15*, 923–933.
19. Chen, T.; Guestrin, C. XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Francisco, CA, USA, 13–17 August **2016**; pp. 785–794.
20. Coqueret, G.; Guida, T. *Machine Learning for Factor Investing*; Chapman and Hall/CRC: Boca Raton, FL, USA, **2020**.
21. Gu, S.; Kelly, B.; Xiu, D. Autoencoder asset pricing models. *J. Econom.* **2021**, *222*, 429–450.
22. Dixon, M.F.; Halperin, I.; Bilokon, P. *Machine Learning in Finance: From Theory to Practice*; Springer: Cham, Switzerland, **2020**.
23. Lundberg, S.M.; Lee, S.I. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* **2017**, *30*, 4765–4774.
24. Brown, S.J.; Goetzmann, W.; Ibbotson, R.G.; Ross, S.A. Survivorship bias in performance studies. *Rev. Financ. Stud.* **1992**, *5*, 553–580.

25. Forbes, K.J.; Rigobon, R. No contagion, only interdependence: Measuring stock market comovements. *J. Finance* **2002**, *57*, 2223–2261.
26. Balkema, A.A.; de Haan, L. Residual life time at great age. *Ann. Probab.* **1974**, *2*, 792–804.
27. Pickands, J. Statistical inference using extreme order statistics. *Ann. Stat.* **1975**, *3*, 119–131.
28. Engle, R.F.; Manganelli, S. CAViaR: Conditional autoregressive value at risk by regression quantiles. *J. Bus. Econ. Stat.* **2004**, *22*, 367–381.
29. Kuester, K.; Mittnik, S.; Paolella, M.S. Value-at-risk prediction: A comparison of alternative strategies. *J. Financ. Econom.* **2006**, *4*, 53–89.
30. Longstaff, F.A. The subprime credit crisis and contagion in financial markets. *J. Financ. Econ.* **2010**, *97*, 436–450.
31. Adrian, T.; Brunnermeier, M.K. CoVaR. *Am. Econ. Rev.* **2016**, *106*, 1705–1741.
32. Acharya, V.V.; Pedersen, L.H.; Philippon, T.; Richardson, M. Measuring systemic risk. *Rev. Financ. Stud.* **2017**, *30*, 2–47.
33. López de Prado, M. The 10 reasons most machine learning funds fail. *J. Portf. Manag.* **2018**, *44*, 120–133.
34. Arnott, R.D.; Harvey, C.R.; Kalesnik, V.; Linnainmaa, J.T. Reports of value's death may be greatly exaggerated. *Financ. Anal. J.* **2021**, *77*, 44–67.
35. Ang, A.; Bekaert, G. International asset allocation with regime shifts. *Rev. Financ. Stud.* **2002**, *15*, 1137–1187.
36. Basel Committee on Banking Supervision. Regulatory Consistency Assessment Programme (RCAP): Analysis of Risk-Weighted Assets for Market Risk; Bank for International Settlements: Basel, Switzerland, 2013.
37. Board of Governors of the Federal Reserve System. Supervisory Guidance on Model Risk Management (SR 11-7); Federal Reserve: Washington, DC, USA, 2011.
38. Kirilenko, A.; Kyle, A.S.; Samadi, M.; Tuzun, T. The Flash Crash: High-frequency trading in an electronic market. *J. Finance* **2017**, *72*, 967–998.
39. Dunis, C.L.; Laws, J.; Sermpinis, G. Modelling and trading the EUR/USD exchange rate at the ECB fixing. *Eur. J. Finance* **2010**, *16*, 541–560.
40. Jorion, P. Value at Risk: The New Benchmark for Managing Financial Risk, 3rd ed.; McGraw-Hill: New York, NY, USA, 2007.
41. Fischer, T.; Krauss, C. Deep learning with long short-term memory networks for financial market predictions. *Eur. J. Oper. Res.* **2018**, *270*, 654–669.
42. Zhang, Z.; Zohren, S.; Roberts, S. Deep learning for portfolio optimization. *J. Financ. Data Sci.* **2020**, *2*, 8–20.
43. Brownlees, C.; Engle, R.F. SRISK: A conditional capital shortfall measure of systemic risk. *Rev. Financ. Stud.* **2017**, *30*, 48–79.
44. Guidolin, M.; Timmermann, A. Asset allocation under multivariate regime switching. *J. Econ. Dyn. Control* **2007**, *31*, 3503–3544.
45. Baele, L. Volatility spillover effects in European equity markets. *J. Financ. Quant. Anal.* **2005**, *40*, 373–401.